UNIVERSITY OF LONDON

MSc EXAMINATION 2025

For Internal Students of
Royal Holloway

DO NOT TURN OVER UNTIL TOLD TO BEGIN

IY5606: Secure and Trustworthy Embedded Technologies in Autonomous Systems

Time Allowed: **TWO** hours

Attempt **three** questions

Please answer each question on a separate page
No credit will be given for attempting any further questions

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Question 1

Smart card technology

(a) Briefly explain the key features and benefits of smart cards in comparison to magnetic stripe cards and memory cards. [4]

(b) Define RFID technology and describe its two primary types, active and passive, along with examples of their common applications. [4]

(c) Explain the role of ISO/IEC 7816 and ISO/IEC 14443 standards in the context of smart cards and RFID. [2]

(d) Describe how a passive RFID device communicates with a reader and obtains power for its operation. [5]

(e) Outline the primary steps in a smart card personalization process and discuss the security considerations involved in this stage. [5]

(f) Smart cards are being almost exclusively used in Point-of-Sale (POS) transactions. Explain how contactless smart card communication protocols, such as NFC, ensure secure transactions. [5]

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Question 2

Payment Systems and the Europay-MasterCard-Visa (EMV) Standards

(a) Briefly explain the role of the EMV standard in modern payment systems and its importance for secure transactions. [4]

(b) Describe the difference between Static Data Authentication (SDA) and Dynamic Data Authentication (DDA) in EMV. [4]

(c) Discuss the key challenges of implementing EMV contactless transactions and how they are mitigated. [5]

(d) Explain the purpose of a Transaction Certificate (TC) and how it supports fraud prevention in EMV payment systems. [5]

(e) Tokenization is often used alongside EMV. Define tokenization and describe how it enhances payment security. [4]

(f) Discuss the role of Two-Factor Authentication (2FA) in online payment systems and how it complements EMV in card-not-present (CNP) transactions. [3]

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Question 3

Secure Mobile Telecommunications

(a) Explain the role of the SIM and USIM in mobile telecommunications, highlighting their key differences in terms of functionality and security features. [5]

(b) Discuss the role of SIM and USIM in enabling secure roaming between networks. [5]

(c) Discuss the security requirements of IoT devices in telecommunications networks and how USIM-based authentication can address these requirements. [5]

(d) Explain the specific challenges faced by automotive systems in ensuring secure communication and how telecommunication standards can mitigate these risks. [5]

(e) Highlight the vulnerabilities in GSM security exposed by attacks such as those on the COMP128 algorithm, and explain how these have been addressed in USIM. [3]

(f) Describe the role of Over-the-Air (OTA) updates in SIM/USIM systems and discuss their security implications. [2]

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Question 4

Smart Card Security and Attacks

(a) Outline the primary goals of smart card security and describe two types of threats faced during card usage. [5]

(b) Explain the difference between invasive and non-invasive attacks on smart cards, providing an example of each. [4]

(c) Discuss the principle of "side-channel attacks" and how countermeasures such as masking and balancing address these threats. [4]

(d) Describe the process and significance of glitch attacks, and outline techniques to detect and prevent them. [5]

(e) Highlight the vulnerabilities of Java Card platforms and suggest strategies to mitigate security risks. [4]

(f) Explain the role of tamper resistance in smart card design and describe how protective layers and active shields enhance security. [3]

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Question 5

Imagine you are an information security consultant specialising in smart cards, tokens, security, and applications. You have been assigned a project that requires the provision of expert advisory services for a major city council.

A city plans to modernize its public transport payment system by introducing NFC-enabled handsets for passengers. The system will allow passengers to use their smartphones as virtual transport cards to tap in and out of buses, trains, and trams. The solution must ensure secure transactions, prevent fare evasion, and provide a seamless user experience. Additionally, it should integrate with existing transport ticketing systems.

Your task is to design and analyze the security of the proposed NFC handset-based transport payment system by addressing the following questions:

(a) System Overview: Describe the architecture of the NFC payment system, including the role of handsets, NFC readers, and backend servers. [5]

(b) Security Features of NFC Handsets: Explain how secure elements (e.g., Trusted Execution Environments) in handsets can protect payment credentials. [5]

(c) Security Requirements: Identify key security requirements to ensure system integrity and prevent fare evasion. [5]

(d) Threat Analysis and Countermeasures: Analyze potential security threats and provide countermeasures. [5]

(e) System Integration and Monitoring: Discuss how the NFC payment system can integrate with existing ticketing systems and monitor transactions for anomalies. [5]

END

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